

Section 1: Gravity and Tidal Forces

1. **The periodic rise and fall of sea levels are called:**
 - A) Currents
 - **B) Tides**
 - C) Waves
 - D) Tsunamis
2. **The primary cause of ocean tides on Earth is the gravitational pull of:**
 - A) The Sun
 - **B) The Moon**
 - C) Mars
 - D) Venus
3. **Tidal force is a secondary effect of the force of:**
 - A) Magnetism
 - B) Friction
 - **C) Gravity**
 - D) Inertia
4. **High tides occur on the side of the Earth facing the Moon and also:**
 - **A) On the exactly opposite side**
 - B) At the North Pole only
 - C) At the South Pole only
 - D) Nowhere else
5. **When the Sun, Moon, and Earth are aligned in a straight line, the tides are called:**
 - A) Neap Tides
 - **B) Spring Tides**
 - C) Low Tides
 - D) Ebb Tides
6. **Spring tides occur during which moon phases?**
 - **A) New Moon and Full Moon**
 - B) First and Third Quarter
 - C) Crescent Moon
 - D) Every Tuesday
7. **When the Moon and Sun are at right angles (90°) to each other, the tides are called:**

- **A) Neap Tides**
 - B) Spring Tides
 - C) High Tides
 - D) Flood Tides
8. **Neap tides result in:**
- A) The highest high tides
 - **B) The lowest tidal range (small difference between high and low)**
 - C) Constant sea levels
 - D) Underwater earthquakes
9. **The time interval between two successive high tides is approximately:**
- A) 6 hours
 - **B) 12 hours and 25 minutes**
 - C) 24 hours
 - D) 1 hour
10. **Tidal energy can be harnessed to produce:**
- A) Heat
 - **B) Electricity**
 - C) Fossil fuels
 - D) Solar power

Section 2: The Solar System and Stars

11. **The center of our Solar System is:**
- A) The Earth
 - **B) The Sun**
 - C) Jupiter
 - D) The Moon
12. **The Sun is primarily composed of which two gases?**
- **A) Hydrogen and Helium**
 - B) Oxygen and Nitrogen
 - C) Carbon and Iron
 - D) Neon and Argon
13. **The process by which the Sun produces energy is:**
- A) Nuclear Fission
 - **B) Nuclear Fusion**
 - C) Combustion

- D) Photosynthesis
- 14. **Which planet is known as the "Red Planet"?**
 - A) Venus
 - **B) Mars**
 - C) Saturn
 - D) Mercury
- 15. **The largest planet in our solar system is:**
 - A) Neptune
 - B) Earth
 - **C) Jupiter**
 - D) Saturn
- 16. **The path a planet takes around the Sun is called its:**
 - **A) Orbit**
 - B) Axis
 - C) Rotation
 - D) Equator
- 17. **A "Light Year" is a unit of:**
 - A) Time
 - **B) Distance**
 - C) Speed
 - D) Intensity
- 18. **The Earth rotates on its axis from:**
 - A) East to West
 - **B) West to East**
 - C) North to South
 - D) Top to Bottom
- 19. **The change of seasons on Earth is caused by:**
 - A) The Earth's rotation
 - **B) The tilt of the Earth's axis and its revolution**
 - C) The distance from the Moon
 - D) Solar flares
- 20. **Which planet is closest to the Sun?**
 - **A) Mercury**
 - B) Venus
 - C) Earth
 - D) Mars

Section 3: Rapid Fire Review (21-100)

21. **Natural satellite of Earth?** (Ans: **The Moon**)
22. **Shape of planetary orbits?** (Ans: **Elliptical**)
23. **Hottest planet in the solar system?** (Ans: **Venus**)
24. **Planet with the most prominent rings?** (Ans: **Saturn**)
25. **Time taken for Earth to complete one revolution?** (Ans: **365.25 days**)
26. **Time taken for Earth to rotate once?** (Ans: **24 hours**)
27. **What is an 'Eclipse'?** (Ans: **Obscuring of light from one celestial body by another**)
28. **Solar eclipse occurs during?** (Ans: **New Moon**)
29. **Lunar eclipse occurs during?** (Ans: **Full Moon**)
30. **What is a 'Galaxy'?** (Ans: **A massive system of stars, gas, and dust**)
31. **Name of our galaxy?** (Ans: **The Milky Way**)
32. **What is a 'Supernova'?** (Ans: **A powerful and luminous stellar explosion**)
33. **What is a 'Black Hole'?** (Ans: **A region of space with gravity so strong nothing can escape**)
34. **Does the Moon have its own light?** (Ans: **No, it reflects sunlight**)
35. **Smallest planet?** (Ans: **Mercury**)
36. **Earth's 'Twin' planet?** (Ans: **Venus**)
37. **Planet farthest from the Sun?** (Ans: **Neptune**)
38. **What is 'Gravity'?** (Ans: **Force that attracts objects toward each other**)
39. **First human to walk on the moon?** (Ans: **Neil Armstrong**)
40. **What is a 'Comet'?** (Ans: **Icy body that releases gas/dust, forming a tail**)
41. **What is an 'Asteroid'?** (Ans: **Rocky object orbiting the Sun, smaller than a planet**)
42. **Where is the Asteroid Belt?** (Ans: **Between Mars and Jupiter**)
43. **What are 'Meteoroids'?** (Ans: **Small rocks in space**)
44. **What is a 'Shooting Star'?** (Ans: **A meteor burning up in Earth's atmosphere**)
45. **What is a 'Meteorite'?** (Ans: **A meteor that survives the atmosphere and hits the ground**)

46. **Instrument to see distant stars?** (Ans: **Telescope**)
47. **What is a 'Constellation'?** (Ans: **A group of stars forming a pattern**)
48. **Closest star to Earth (besides the Sun)?** (Ans: **Proxima Centauri**)
49. **What is 'Escape Velocity'?** (Ans: **Speed needed to break free from a planet's gravity**)
50. **Distance of the Sun from Earth?** (Ans: **Approx. 150 million km**)
51. **Number of planets in our solar system?** (Ans: **8**)
52. **Is Pluto currently a major planet?** (Ans: **No, it is a Dwarf Planet**)
53. **What is 'Absolute Zero'?** (Ans: **Lowest possible temperature, -273.15°C**)
54. **What is 'Orbital Velocity'?** (Ans: **Speed needed to maintain an orbit**)
55. **Does the Moon have an atmosphere?** (Ans: **No, effectively a vacuum**)
56. **Gravity on the Moon compared to Earth?** (Ans: **1/6th of Earth's gravity**)
57. **Outer layer of the Sun?** (Ans: **Corona**)
58. **Visible surface of the Sun?** (Ans: **Photosphere**)
59. **What are 'Sunspots'?** (Ans: **Cooler, darker spots on the Sun's surface**)
60. **What are 'Tidal Bulges'?** (Ans: **Swelling of water caused by gravity**)
61. **What is 'Centrifugal Force' in tides?** (Ans: **Force pushing water outward on the opposite side of the Moon**)
62. **Number of high tides in 24 hours?** (Ans: **2**)
63. **What is 'Apogee'?** (Ans: **Point where Moon is farthest from Earth**)
64. **What is 'Perigee'?** (Ans: **Point where Moon is closest to Earth**)
65. **Do tides occur in lakes?** (Ans: **Yes, but they are too small to notice**)
66. **What is 'Equinox'?** (Ans: **When day and night are of equal length**)

67. **When is the Summer Solstice?** (Ans: **June 21 - Longest day**)
68. **When is the Winter Solstice?** (Ans: **Dec 21 - Shortest day**)
69. **What is 'Zenith'?** (Ans: **Point directly overhead**)
70. **What is 'Nadir'?** (Ans: **Point directly below**)
71. **What is 'Cosmology'?** (Ans: **Study of the origin and evolution of the universe**)
72. **What is 'Astronomy'?** (Ans: **Scientific study of celestial objects**)
73. **What is 'Astrology'?** (Ans: **Non-scientific belief in stars affecting human fate**)
74. **Speed of light?** (Ans: **300,000 km/s**)
75. **Time for sunlight to reach Earth?** (Ans: **Approx. 8 minutes and 20 seconds**)
76. **What color are the hottest stars?** (Ans: **Blue**)
77. **What color are the coolest stars?** (Ans: **Red**)
78. **What is a 'Nebula'?** (Ans: **A cloud of gas and dust where stars are born**)
79. **What is 'Hubble'?** (Ans: **A famous space telescope**)
80. **What determines a star's life span?** (Ans: **Its mass**)
81. **Which planet rotates on its side?** (Ans: **Uranus**)
82. **Does Saturn float in water?** (Ans: **Yes, its density is less than water**)
83. **What is 'Great Red Spot'?** (Ans: **A giant storm on Jupiter**)
84. **Evening or Morning Star?** (Ans: **Venus**)
85. **Gas that makes Neptune blue?** (Ans: **Methane**)
86. **What is 'Aurora'?** (Ans: **Light display in polar regions caused by solar particles**)
87. **What is 'Parallax'?** (Ans: **Shift in position of an object when viewed from different points**)
88. **Unit of astronomical distance?** (Ans: **Astronomical Unit - AU**)
89. **What is 1 AU?** (Ans: **Distance from Earth to the Sun**)
90. **Can sound travel in space?** (Ans: **No**)
91. **Is there a 'Dark Side' of the moon?** (Ans: **No, but there is a 'Far Side' we don't see from Earth**)
92. **Why do we see only one side of the Moon?** (Ans: **Synchronous rotation**)

93. **What is 'Ebb'? (Ans: Receding tide)**
94. **What is 'Flood'? (Ans: Incoming tide)**
95. **Does the Sun cause tides? (Ans: Yes, but less than the Moon)**
96. **What is 'Magnitude' of a star? (Ans: Its brightness)**
97. **Who proposed the Heliocentric model? (Ans: Copernicus)**
98. **Who discovered Jupiter's moons? (Ans: Galileo)**
99. **Is our Sun a medium-sized star? (Ans: Yes, a Yellow Dwarf)**
100. **The ultimate fate of the Sun? (Ans: It will become a White Dwarf)**